Mohammadhossein Sabzalian

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Summary

- Final-year Electrical Engineering student seeking a master's in digital systems, embedded systems and data science.
- GPA 16.88/20 (department average 16.49); 19.30/20 in the most recent semester.
- Four semesters of teaching assistance; a teaching-support platform adopted by a second university.

Education

2023–2027 **Sharif University of Technology, Tehran, Iran**
B.Sc., Electrical Engineering (Digital Systems focus) • expected Aug 2027

- **Cumulative GPA: 16.88/20**. Above the department average (16.49) and university average (16.09) for this cohort, as printed on the Sharif transcript.
- Most recent semester (Spring 2026): **19.30/20** over 19 units.
- **National rank 134** among about 150,000 candidates in the Mathematics and Physics group of the Iranian University Entrance Exam (Konkur).
- **Relevant coursework (grade/20):** ASIC/FPGA System Design (19.4); Embedded Systems (19.8); Foundations of Data Science (19.3); Computer Interfacing, a graduate course (19.0); Logic Circuits and Digital Systems with lab (18.2); Microprocessor Systems Design (18.1).

Course Projects

Spring 2026 **Smart Surveillance System on an Embedded Linux Board** source
Embedded Systems course, Dr. Gholampour • Orange Pi Zero Plus 2 (H5), C, Python

- **System:** Individual project built around an on-board person-detection system in which a Haar-cascade detector feeds an HTTPS dashboard, a REST API and an MQTT/e-mail alert daemon.
- **Reliability:** Added a software watchdog that restarts the detector when the camera feed stalls, hysteresis-based thermal throttling and an event log.
- **Evaluation:** Measured boot time, detection accuracy under varied lighting, concurrent load and temperature with and without a fan; documented in a technical report.

Spring 2026 **OTP Memory Controller (RTL design, synthesis and verification)** source
ASIC/FPGA System Design course, Dr. Baharvand • Verilog, Cadence Genus and Xcelium

- **Design:** Co-designed a power-on controller: an FSM reads a 256-bit one-time-programmable array, applies two hardware patch overrides to faulty bits, checks a CRC-16 and latches the verified calibration values into output registers.
- **Verification and flow:** Verified the design with a self-checking testbench, then ran synthesis and gate-level simulation with SDF timing back-annotation.
- **Team:** Team of 4 with a rotating weekly lead; led the first week, planning the work and assigning tasks.

Spring 2025 **MIPS32 Processor Implementation on FPGA**
Computer Architecture lab, Dr. Movahedin • Verilog, MIPS32 assembly, Xilinx Zynq-7010

- **Design:** Implemented single-cycle, multi-cycle and pipelined MIPS32 processors.
- **Validation:** Ran a custom multi-cycle CPU on an FPGA board and wrote a working calculator program for it.

2026–present **SUT Music: Data Platform and Recommendation Engine** source
Foundations of Data Science course, Dr. Salavati • Python, scikit-learn, PostgreSQL, FastAPI, React

- **Data:** Led the scraping and cleaning of a Telegram music community's history into a database of 14,800+ tracks and 86,900+ reactions.
- **Recommenders:** Built two generations (matrix factorization, then a two-stage artist-to-track ensemble) behind a backend API and a Telegram Web App.
- **Team:** Team of 3; project lead.

Teaching Experience

2024–2026 **Teaching Assistant, Fundamentals of Programming and OOP**
Sharif University of Technology • Dr. Vosoughi Vahdat and Dr. Arasteh • four consecutive semesters

Fall 2025 **Teaching Assistant, Electronics 1**
Sharif University of Technology • Dr. Fakharzadeh, Dr. Sarvari, Dr. Medi, Dr. Besharatirad

Fall 2025 **Lab Assistant, Logic Circuit Lab (FPGA)**
Sharif University of Technology • Dr. Movahedin • supported students designing on Zynq-7010 FPGA boards

Spring 2026 **Teaching Assistant, Theory of Circuits**
Iran University of Science and Technology (IUST)

Industrial Training

Summer 2026 **Trainee, Parsian Zarin (ABLY): MQL5 expert advisors; Excellent Pass**

Software for Teaching and Community Service

2024–present **EE Department Bot Ecosystem and BotOS Framework** source
Lead developer • Python (asyncio), PostgreSQL, Docker, Google Sheets API

- Began with the programming courses I assisted (Fundamentals of Programming, Object-Oriented Programming) and still maintain bots used by the EE department for course-file distribution, assignment grading, and re-judge ticketing, and by two student organisations; the platform has since been adopted by a second university (IUST).
- Designed BotOS, a framework that deploys new bots from declarative JSON configuration in isolated Docker containers.

Technical Skills

- **Digital design:** Verilog, SystemVerilog, RTL design and testbenches; Xilinx Vivado, ModelSim/QuestaSim, Cadence Genus and Xcelium; Xilinx Zynq-7010.
- **Embedded and circuits:** C, assembly (MIPS32), embedded Linux on single-board computers; LTspice, Altium Designer.
- **Software and data:** Python (data analysis, machine learning, backend development), C++, Java, PostgreSQL, SQLite, Docker, Git, CI/CD, Linux.

Languages

Persian (native); English (official test score pending).

Referees

Academic referees available on request; names and contact details provided with the application.